THE HUMAN-STRESS DETECTION IN AND THROUGH SLEEP USING MACHINE LEARNING

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ABSTRACT

In today's fast-paced world, stress is a widespread issue that can negatively affect both physical and mental health. Early identification and monitoring of stress levels are essential for enhancing overall well-being. This project introduces an innovative machine learning method to detect stress by evaluating a person's sleep behavior. It utilizes the Random Forest Classifier, a highly efficient algorithm recognized for its robustness and accuracy in classification problems. The primary aim of this research is to develop a reliable prediction model capable of estimating an individual's stress levels based on various physiological signals recorded during sleep. The dataset comprises several important features, including snoring intensity, breathing rate, body temperature, limb activity, oxygen levels in the blood, eye movements, heart rate, total sleep duration, and stress labels divided into five categories: 0 (Low), 1 (Low-Medium), 2 (Medium), 3 (Medium-High), and 4 (High). By integrating these attributes, the model is trained to detect complex patterns between sleep indicators and stress conditions. The system achieved high classification accuracy, proving the potential of sleep- related data in stress level estimation. This approach can be beneficial for medical diagnostics, personal health tracking, and wellness applications. It empowers individuals to understand how their sleep quality influences stress and take proactive measures to manage it. In the long run, the model can contribute to better mental health support and the promotion of healthier living habits.

Keywords: Machine Learning, Matplotlib, Seaborn, Scikit-learn, NumPy, and Pandas

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1.INTRODUCTION

The study introduces an IoMT-based solution for detecting physiological stress linked to sleep and eating behaviors using a smart device called SaYoPillow. This device monitors real-time biological signals—like heart rate, snoring, respiration, REM sleep, and oxygen levels—to improve sleep quality and reduce stress. A dataset of 348 individuals aged 20–60 was used to train machine learning models. Among them, the Random Forest algorithm achieved a 90% F1-score in stress classification, and EEG data further enhanced accuracy. The system reached 83.4% accuracy across multiple stress levels and 94.6% for level-two stress. Physiological signals such as GSR, ECG, EMG, and respiration were crucial in identifying stress patterns. SVM achieved the best accuracy (73%) for classifying stress into low, moderate, and high levels, while LSTM showed high potential but required larger datasets and computing power.

II. LITERATURE SURVEY

Recent advancements in the Internet of Medical Things (IoMT) have led to the development of methods to identify and manage stress related to eating behaviors and physiological responses. One such innovation is SaYoPillow, a smart pillow designed to monitor and regulate stress levels during sleep. The primary objective of SaYoPillow is to facilitate "Smart Sleeping," aiming to ensure sleep quality aligns with optimal physiological needs. The system monitors various parameters such as heart rate, snoring levels, respiratory rate, total sleep duration, blood oxygen saturation, eye movement (especially during REM sleep), body temperature variations, and limb movements. For instance, snoring volumes above 50 dB are considered to potentially increase stress and other health risks. A healthy breathing rate while asleep is about 15–17 breaths per minute, and the heart rate typically slows down by 5 to 10 beats per minute during sleep. Adults are recommended to get a minimum of seven hours of sleep each night, with 20–25% of that in the Rapid Eye Movement (REM) stage—roughly 90 minutes for optimal health. Oxygen saturation levels falling below 90% are flagged as abnormal, indicating potential physiological stress. A study was conducted on 348 participants, both employed and unemployed, aged between 20 and 60, involved in various occupational and domestic activities. The study employed a machine learning framework to evaluate mental stress at different levels. Among the algorithms used, Random Forest yielded the highest accuracy for stress detection (F1 score of 90%), while Naive Bayes performed best for detecting depression. Another study implemented an EEG-based stress monitoring framework. Stress was artificially induced using the Montreal Imaging Stress Task (MIST), and participants' EEG signals were analyzed using several methods, including t-tests, ROC curves, and Bhattacharyya distance for feature selection. Machine learning techniques such as logistic regression, SVM, and Naive Bayes were employed. The model achieved an overall detection accuracy of 83.4%, with a 94.6% accuracy rate for identifying high-stress levels. Galvanic Skin Response (GSR) was identified as a key physiological marker for stress, reflecting increased skin conductance due to sweat gland activity triggered by the autonomic nervous system. Continuous monitoring of stress was emphasized as critical, especially with the rise of wearable technologies and edge computing for real-time data analysis. Physiological sensor analytics—using devices such as ECG monitors, EMG sensors, and GSR detectors—has become a cornerstone in health and stress monitoring systems. One approach analyzed stress among pilgrims by evaluating their sleep patterns and identifying the most significant sleep-related indicators. Using data from respiration, body temperature, GSR, and body posture sensors (including accelerometers), several classification models were built to detect three stress levels: low, medium, and high. Support Vector Machine (SVM) models demonstrated the highest accuracy at 73%. While Long Short-Term Memory (LSTM) models showed superior performance in some cases, they often require large datasets and significant computational power, making them less suitable for smaller-scale implementations.

III. METHODOLOGY

This study aims to predict human stress by analyzing sleep-related behaviors. To accomplish this, five critical steps are outlined: data gathering, Dataset Creation, Data Preparation, Data Splitting, and Model Selection. A comprehensive description of the study's framework is provided after Fig. 1.

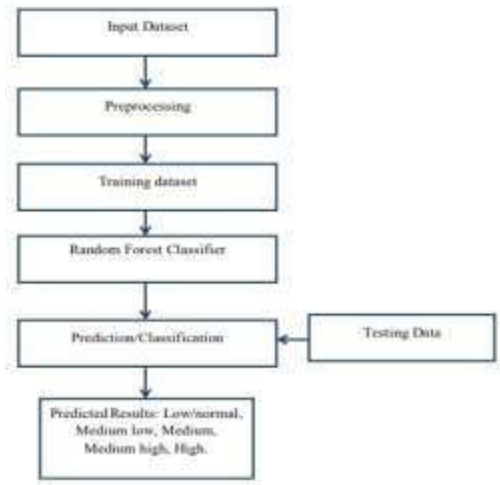


Figure 1: Architecture of the study

3.1 DATA COLLECTION

The dataset used for this study was obtained from Kaggle, a well-known platform for publicly available datasets often used by researchers and developers. It contains 630 records that include both physiological and sleep-related parameters necessary for effective stress level classification. The first stage in developing the Human Stress Detection Based on Sleeping Habits system involved acquiring this dataset. Data collection plays a foundational role in building any machine learning model. The quality and quantity of the collected data directly influence the model's overall performance and accuracy. High-quality data allows the model to learn patterns more effectively, resulting in better predictive capabilities. There are multiple approaches to data collection, such as manual data entry, web scraping, and using pre-existing datasets from reliable sources. In this project, a precompiled dataset from Kaggle was used. It is included in the project directory under the model folder. The dataset primarily consists of numerical features, which are suitable for feeding into machine learning algorithms.

3.2 DATASET

The dataset comprises 630 records, each representing individual sleep and physiological data points. It includes a total of 9 features, which are essential for analyzing and classifying stress levels. Below is a brief explanation of each column: SR (Snoring Range): Measures the decibel level of snoring, which can indicate sleep disturbances and stress. RR (Respiration Rate): Represents the number of breaths per minute, an important marker of relaxation or stress. T (Body Temperature): Indicates the core body temperature during sleep. LM (Limb Movement Rate): Tracks the frequency of limb movement, which may suggest restlessness or discomfort. BO (Blood Oxygen Level): Measures the percentage of oxygen saturation in the blood, with lower levels potentially indicating health risks or stress. REM (Eye Movement): Refers to rapid eye movement, a phase of sleep critical for mental restoration. SR1 (Sleep Duration): Denotes the total number of hours the individual slept. HR (Heart Rate): Captures the average heart rate during sleep. SL (Stress Level): This is the target variable with five categories:

0: Low/Normal

1: Medium-Low

2: Medium

3: Medium-High

4: High

These features serve as input variables for the machine learning model to predict the level of stress based on sleeping patterns and physiological behavior.

3.3 DATA PREPARATION

The preparation of data started with a comprehensive cleaning process that involved eliminating duplicate records, resolving data inconsistencies, and managing any missing or incomplete values. Numerical features were scaled to a standard range to maintain uniformity, and appropriate data type conversions were applied where needed. To minimize potential bias resulting from the sequence in which the data were collected or arranged, the entire dataset was randomized. To better understand the dataset and uncover useful patterns, Exploratory Data Analysis (EDA) was performed using visualization tools. These visual insights helped identify correlations among variables and spot any imbalanced class distributions that could influence model outcomes. Tools like correlation matrices and distribution graphs were utilized for this analysis. Once the data was fully preprocessed, it was divided into two parts: a training set and a testing set. This separation ensures that the machine learning model can be trained effectively while allowing its performance to be validated on unseen data, thus ensuring better reliability and generalization of the results.

3.4 SPLITTING THE DATASET

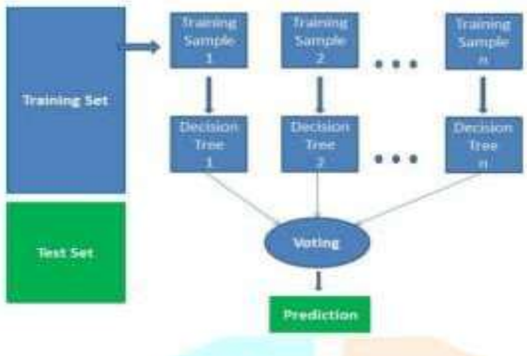
In this module, the image dataset will be divided into training and testing sets. Split the dataset into Train and Test. 80% train data and 20% test data. This will be done to train the model on a subset of the data, validate the model's performance, and test the model on unseen data to evaluate its accuracy. Split the dataset into train and test. 80% train data and 20% test data.

3.5 MODEL SELECTION

We used 1. Random Forest Classifier and 2. Decision Tree machine learning algorithms, and we got an accuracy of 97.62% and 96.83% on the test set, so we implemented these algorithms.

1.The Random Forest Algorithm

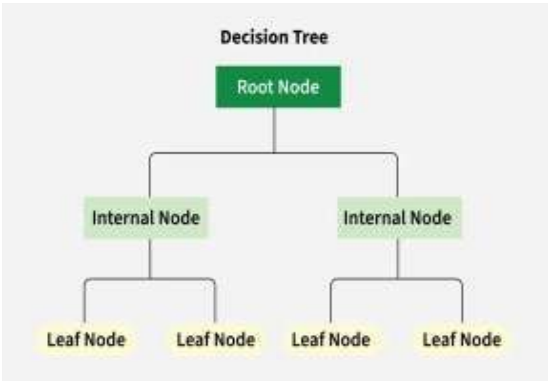
The Random Forest algorithm is an ensemble learning method that creates multiple decision trees and combines their outputs to generate a final prediction. Each tree is trained on a randomly selected subset of the data and uses a random set of features for decision-making at each split. The final result is determined by majority voting for classification tasks or averaging for regression tasks. This collective approach boosts accuracy and mitigates overfitting, making Random Forest highly effective for handling complex and large datasets. One of the key advantages of Random Forest is its ability to assess the importance of different features. By measuring how much each feature contributes to reducing prediction error, the model can pinpoint the most relevant variables, optimizing the performance and interpretability of the model. Additionally, Random Forest is known for its robustness to outliers and noise. The randomization of both data and features ensures that anomalies have minimal impact on the final predictions. This makes it particularly useful for real-world applications, such as analyzing physiological data in stress detection, where inconsistencies and noise are common. Despite its many strengths, the model can be computationally demanding, particularly when using large datasets or an extensive number of trees. Nevertheless, its capability to process noisy, high-dimensional data, along with its accuracy and adaptability, makes Random Forest a popular and reliable choice for various machine learning tasks.



1.Random Forest Algorithm Diagram.

2.The Decision Tree Algorithm

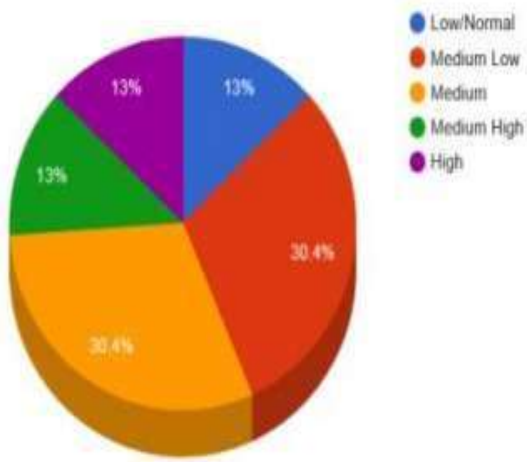
A decision tree is a supervised machine learning algorithm used for solving both classification and regression problems. It works by splitting data into subsets based on feature values, creating a tree-like model where each internal node represents a decision rule, each branch shows an outcome of that rule, and each leaf node signifies a final prediction—either a class label or a numerical value. The algorithm selects the most informative feature at each step using criteria like Gini impurity, Information Gain, or Mean Squared Error. It recursively partitions the data to form purer subsets, continuing until a stopping condition is reached, such as a maximum depth or minimum number of samples. Decision trees are widely favored for their interpretability. They allow users to clearly understand the flow of decisions, making them useful in domains requiring transparent models like healthcare and finance. The visual structure helps in tracing how the model reaches a particular outcome. They can handle both categorical and continuous variables, detect non-linear relationships, and require minimal data preprocessing. However, decision trees are prone to overfitting, especially if the tree grows too deep. To improve generalization, techniques like pruning or limiting tree depth are used. In applications like stress detection through sleep data, decision trees can identify which physiological features—such as snoring intensity or heart rate—most strongly indicate stress. While simpler than ensemble models, they offer fast, explainable, and effective predictions

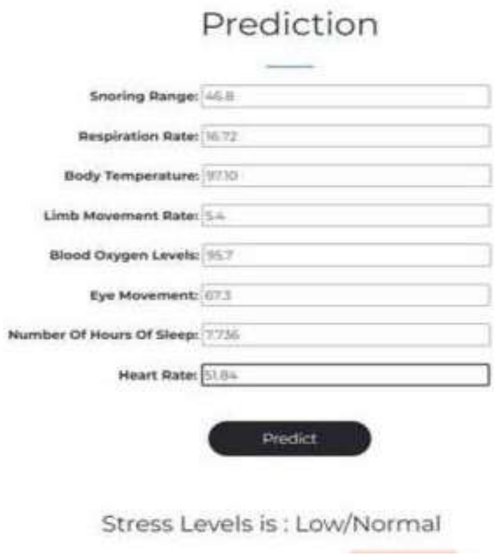


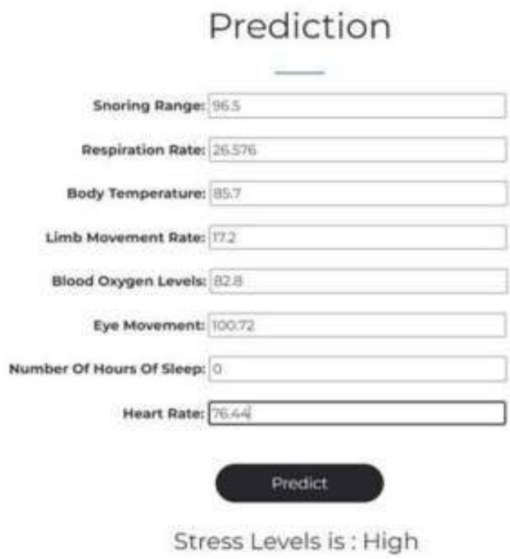
2. Decision Tree Algorithm Diagram.

IV. RESULTS AND DISCUSSION

In machine learning, prediction involves utilizing a model that has been trained on historical data to generate outputs for new and previously unseen inputs. This process allows the model to apply the patterns and relationships it identified during training to make informed decisions or estimations. The model is provided with new input data that shares structural similarity with the training dataset in terms of features and format. Before predictions are made, the input data undergoes preprocessing. This includes cleaning, normalization, feature selection, or transformation to ensure the data is in the appropriate form for the model to interpret. Proper preprocessing is critical, as it ensures consistency with the data used during the training phase and significantly impacts the quality of the predictions. Once the data is prepared, the model utilizes its learned parameters and internal logic to generate predictions. For classification problems, the output may be discrete class labels, whereas in regression tasks, the model produces continuous numeric values. Following the prediction, the performance of the model is thoroughly evaluated using standard metrics. In classification, these may include accuracy, precision, recall, and the F1-score, which collectively offer insight into the correctness and robustness of the predictions. For regression problems, evaluation may involve metrics like mean squared error (MSE) or root mean squared error (RMSE) to assess the closeness of predicted values to actual targets. These performance metrics help determine whether the model is overfitting, underfitting, or successfully generalizing to unseen data. Visual tools such as confusion matrices, ROC curves, or error distribution plots may also be used for a deeper understanding of the model's behavior. Once validated and fine-tuned, the model becomes suitable for practical deployment. It can then be integrated into real-world systems where it assists in decision-making, monitoring, classification, prediction, or automation, based on newly incoming data.







CONCLUSION

This study demonstrates the effective application of machine learning techniques for predicting outcomes based on patterns learned from historical data. By training the model on labeled datasets and validating its performance on unseen inputs, reliable and accurate predictions can be achieved. The preprocessing of data plays a crucial role in maintaining consistency and ensuring that the model receives clean and relevant input. Through the use of various evaluation metrics, the model's strengths and weaknesses are thoroughly analyzed, enabling the detection of overfitting, underfitting, or potential improvements. Once validated, the trained model proves to be a valuable tool for real-world applications, providing timely and data-driven insights for decision-making and automation. Overall, the integration of such models into practical environments highlights the potential of machine learning to solve complex problems efficiently and with high accuracy.

RELATED WORK

In the existing literature, researchers have proposed various stress detection systems leveraging Internet of Medical Things (IoMT) devices such as SaYoPillow, which monitor physiological signals including heart rate, snoring, respiration, and body temperature during sleep. These studies employed machine learning algorithms like SVM, Naive Bayes, and LSTM, often using EEG and GSR data to classify stress levels with considerable accuracy. While these approaches focused on hardware-based sensor inputs and controlled lab settings, my project took a data-driven software approach using a publicly available dataset from Kaggle. I implemented machine learning models—specifically, Random Forest and Decision Tree classifiers—to detect human stress through features like snoring range, respiration rate, sleep duration, heart rate, and oxygen saturation. My model achieved high accuracy (up to 97.62%) in classifying stress into five levels, based on sleep behavior patterns. Additionally, to make the solution user-friendly and accessible, I developed a web page that integrates the trained model, allowing users to input sleep-related parameters and receive real-time stress predictions. This interface also visualizes the feature importance and classification results, bridging the gap between technical analysis and practical usability. Through this, my work contributes not only a high-performing model but also a complete, interactive solution for stress detection.

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